KAIZEN MATHS

Lesson starter

Momentum and Impulse

Conservation of Momentum

Name _____

S1

A particle of mass 2 kg moving to the right at 10 m s^{-1} collides with a particle of mass 3 kg moving to the right at 2 m s^{-1}. They stick together after the collision. Find their common speed.

S2

Two particles of masses 6 kg and 8 kg move on the same line. Before collision their velocities are 10 m s^{-1} to the right and 0 m s^{-1}. After collision, the 6 kg particle has velocity 3 m s^{-1} to the right. Find the velocity of the 8 kg particle after collision.

S3

A body at rest explodes into two parts of masses 3 kg and 5 kg. The 3 kg part moves to the right at 8 m s^{-1}. Find the velocity of the 5 kg part.

S4

A 0.05 kg pellet is fired horizontally to the right from a 3 kg trolley initially at rest. The pellet leaves at 150 m s^{-1}. Find the recoil velocity of the trolley.

Show your working

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MEC-503C-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Momentum and Impulse

Conservation of Momentum

Name _____

EA1

A particle of mass 2 kg moving to the right at 12 m s^{-1} collides with a particle of mass 3 kg moving to the right at 1 m s^{-1}. They stick together after the collision. Find their common speed.

EA2

Two particles of masses 8 kg and 8 kg move on the same line. Before collision their velocities are 5 m s^{-1} to the right and 3 m s^{-1} to the left. After collision, the 8 kg particle has velocity 2 m s^{-1} to the right. Find the velocity of the 8 kg particle after collision.

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KAIZEN MATHS

Exit ticket

Momentum and Impulse

Conservation of Momentum

Name _____

EB1

A body at rest explodes into two parts of masses 1 kg and 6 kg. The 1 kg part moves to the right at 4 m s^{-1}. Find the velocity of the 6 kg part.

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KAIZEN MATHS

Exit ticket

Momentum and Impulse

Conservation of Momentum

Name _____

EA1

A particle of mass 2 kg moving to the right at 12 m s^{-1} collides with a particle of mass 3 kg moving to the right at 1 m s^{-1}. They stick together after the collision. Find their common speed.

EA2

Two particles of masses 8 kg and 8 kg move on the same line. Before collision their velocities are 5 m s^{-1} to the right and 3 m s^{-1} to the left. After collision, the 8 kg particle has velocity 2 m s^{-1} to the right. Find the velocity of the 8 kg particle after collision.

MEC-503C-L2 · EA

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KAIZEN MATHS

Exit ticket

Momentum and Impulse

Conservation of Momentum

Name _____

EB1

A body at rest explodes into two parts of masses 1 kg and 6 kg. The 1 kg part moves to the right at 4 m s^{-1}. Find the velocity of the 6 kg part.

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Teacher answers & working

Momentum and Impulse · Conservation of Momentum

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S1

A particle of mass 2 kg moving to the right at 10 m s^{-1} collides with a particle of mass 3 kg moving to the right at 2 m s^{-1}. They stick together after the collision. Find their common speed.

Answer:

5.2 m s^{-1} to the right

Working:

Take right as positive.

Because the particles stick together, they have one common final velocity v.

Momentum before = 2(10) + 3(2)

Momentum after = (2 + 3)v

Use conservation of momentum.

$2(10) + 3(2) = (2 + 3)v$

$26 = 5v$

$v = 26 \div 5 = 5.2$ m s^{-1}

The common speed is 5.2 m s^{-1} to the right.

S2

Two particles of masses 6 kg and 8 kg move on the same line. Before collision their velocities are 10 m s^{-1} to the right and 0 m s^{-1}. After collision, the 6 kg particle has velocity 3 m s^{-1} to the right. Find the velocity of the 8 kg particle after collision.

Answer:

5.25 m s^{-1} to the right

Working:

Take right as positive.

Before: $u_1 = 10$, $u_2 = 0$

After: $v_1 = 3$, v_2 is unknown

Write total momentum before and after.

Momentum before = 6(10) + 8(0) = 60

Momentum after = $6(3) + 8v_2$

Use conservation of momentum.

$60 = 6(3) + 8v_2$

$60 = 18 + 8v_2$

$8v_2 = 42$

$v_2 = 5.25$ m s^{-1}

The velocity is 5.25 m s^{-1} to the right.

Teacher answers & working

Momentum and Impulse · Conservation of Momentum

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S3

A body at rest explodes into two parts of masses 3 kg and 5 kg. The 3 kg part moves to the right at 8 m s^{-1}. Find the velocity of the 5 kg part.

Answer:

4.8 m s^{-1} to the left

Working:

Take right as positive.

The body is initially at rest, so the total momentum before the explosion is 0.

Momentum before = 0

Momentum after = 3(8) + 5v

Use conservation of momentum.

$0 = 3(8) + 5v$

$0 = 24 + 5v$

$5v = -24$

$v = -4.8$ m s^{-1}

The velocity is 4.8 m s^{-1} to the left.

S4

A 0.05 kg pellet is fired horizontally to the right from a 3 kg trolley initially at rest. The pellet leaves at 150 m s^{-1}. Find the recoil velocity of the trolley.

Answer:

2.5 m s^{-1} to the left

Working:

Take right as positive.

Initially the pellet and trolley are at rest, so the total momentum before firing is 0.

Momentum before = 0

Momentum after = (0.05)(150) + 3v

Use conservation of momentum for the pellet-trolley system.

$0 = 7.5 + 3v$

$3v = -7.5$

$v = -2.5$ m s^{-1}

The trolley recoils at 2.5 m s^{-1} to the left.

Teacher answers & working

Momentum and Impulse · Conservation of Momentum

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EA1

A particle of mass 2 kg moving to the right at 12 m s^{-1} collides with a particle of mass 3 kg moving to the right at 1 m s^{-1}. They stick together after the collision. Find their common speed.

Answer:

5.4 m s^{-1} to the right

Working:

Take right as positive.

Because the particles stick together, they have one common final velocity v.

Momentum before = 2(12) + 3(1)

Momentum after = (2 + 3)v

Use conservation of momentum.

$2(12) + 3(1) = (2 + 3)v$

$27 = 5v$

$v = 27 \div 5 = 5.4$ m s^{-1}

The common speed is 5.4 m s^{-1} to the right.

EA2

Two particles of masses 8 kg and 8 kg move on the same line. Before collision their velocities are 5 m s^{-1} to the right and 3 m s^{-1} to the left. After collision, the 8 kg particle has velocity 2 m s^{-1} to the right. Find the velocity of the 8 kg particle after collision.

Answer:

0 m s^{-1}

Working:

Take right as positive.

Before: $u_1 = 5$, $u_2 = -3$

After: $v_1 = 2$, v_2 is unknown

Write total momentum before and after.

Momentum before = 8(5) + 8(-3) = 16

Momentum after = $8(2) + 8v_2$

Use conservation of momentum.

$16 = 8(2) + 8v_2$

$16 = 16 + 8v_2$

$8v_2 = 0$

$v_2 = 0$ m s^{-1}

The velocity is 0 m s^{-1}.

Teacher answers & working

Momentum and Impulse · Conservation of Momentum

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EB1

A body at rest explodes into two parts of masses 1 kg and 6 kg. The 1 kg part moves to the right at 4 m s^{-1}. Find the velocity of the 6 kg part.

Answer:

0.667 m s^{-1} to the left

Working:

Take right as positive.

The body is initially at rest, so the total momentum before the explosion is 0.

Momentum before = 0

Momentum after = 1(4) + 6v

Use conservation of momentum.

$0 = 1(4) + 6v$

$0 = 4 + 6v$

$6v = -4$

$v = -0.667$ m s^{-1}

The velocity is 0.667 m s^{-1} to the left.